

Chapter 9 Work, energy and power

Section 9.1

Worked example: Try yourself 9.1.1

CALCULATING WORK

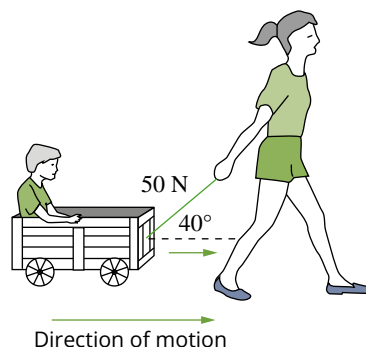
A person pushes a heavy wardrobe from one room to another by applying a force of 50 N for a distance of 5 m. Calculate the amount of work done.

Thinking	Working
Recall the definition of work.	$W = Fs$
Substitute in the values for this situation.	$W = 50 \times 5$
Solve the problem, giving an answer with appropriate units.	$W = 250 \text{ J}$

Worked example: Try yourself 9.1.2

WORK WITH FORCE AND DISPLACEMENT AT AN ANGLE

A girl pulls her brother along in a trolley for a distance of 30 m, as shown. Calculate the work done on the box. Give your answer correct to three significant figures.

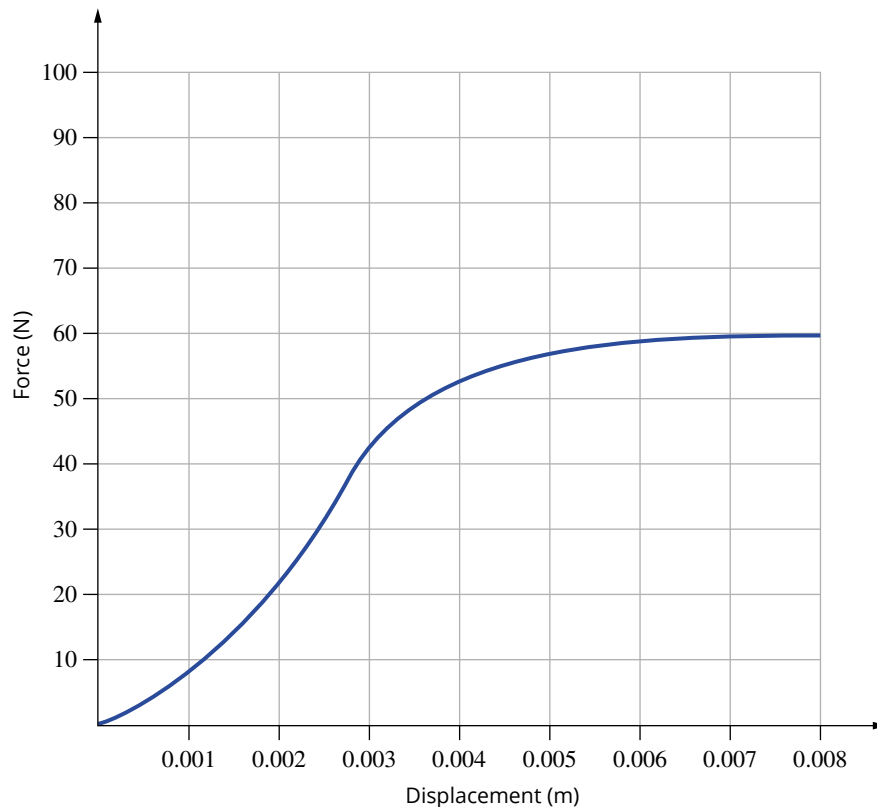


Thinking	Working
Determine values for F , s and θ .	$F = 50 \text{ N}$ $s = 30 \text{ m}$ $\theta = 40^\circ$
Recall the work equation.	$W = Fs \cos \theta$
Substitute values into the work equation.	$W = 50 \times 30 \times \cos 40^\circ$
State the answer with the correct units.	$W = 1150 \text{ J}$

Worked example: Try yourself 9.1.3

WORK FROM THE AREA UNDER A FORCE–DISPLACEMENT GRAPH

While jogging, a person's shoes compress by an average of 3 mm with each step. Use the force–displacement graph for a sports shoe to estimate how much work is done on the shoe with each step. Give your answer to the nearest 0.01 J.



Thinking	Working
Calculate the work value of each grid square.	The dimensions of a grid square are: Force: 10 N, displacement: 0.001 m Area of 1 square = $10 \times 0.001 = 0.01 \text{ J}$
Count the number of grid squares under the curve up to a distance of 3 mm or 0.003 m. Only count grid squares that are more than half under the curve. If the curve cuts a square in half, count every second one.	Number of squares = 5
Multiply the number of grid squares under the curve by the work value of each grid square.	$W = 5 \times 0.01 = 0.05 \text{ J}$

Section 9.1 Review

- $W = Fs = 500 \times 20 = 10\,000 \text{ J}$
- The person exerts a force on the wall but the wall has no displacement ($s = 0$), so no work is done.
- $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 0.015 \times 200 = 1.5 \text{ J}$
- $W = Fs$
 $2700 = F \times 150$
 $F = 18 \text{ N}$

- 5 $W = Fs \cos \theta$
 $= 80 \times 5.0 \times \cos 40^\circ$
 $= 310 \text{ J}$
- 6 The equation $W = Fs$ applies to situations where the applied force is constant. Since a spring obeys Hooke's law, the force required to compress a spring is not constant.
- 7 Since the box does not move, no work is done.
- 8 The area under the graph is a triangle.
 $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 6 \times 10^{-3} \times 1200 = 3.6 \text{ J}$
- 9 Work done is the area under the force-displacement graph. This can be calculated by approximating the area to a geometrical shape or by counting squares.
 $1 \text{ square} = 20 \text{ N} \times 10 \text{ mm} = 20 \text{ N} \times 0.01 \text{ m} = 0.2 \text{ J}$
 W_A is approximately $15 \text{ squares} \times 0.2 = 3.0 \text{ J}$
 W_B is approximately $12 \text{ squares} \times 0.2 = 2.4 \text{ J}$
 W_C is approximately $5 \text{ squares} \times 0.2 = 1.0 \text{ J}$
- 10 a $A = \frac{1}{2}bh = \frac{1}{2} \times 0.04 \times 400 = 8 \text{ J}$
b $A = \frac{1}{2}bh = \frac{1}{2} \times 0.03 \times 300 = 4.5 \text{ J}$
c As the basketball bounces, some energy is lost as heat and sound so the work when the ball rebounds is less than the work done when the ball compresses.

Section 9.2 Kinetic energy

Worked example: Try yourself 9.2.1

CALCULATING KINETIC ENERGY

A person crossing the street is walking at 5.0 km h^{-1} . If the person has a mass of 80 kg , calculate their kinetic energy. Give your answers correct to two significant figures.	
Thinking	Working
Convert the person's speed to m s^{-1} .	$5 \text{ km h}^{-1} = \frac{5 \text{ km}}{1 \text{ h}} = \frac{5000 \text{ m}}{3600 \text{ s}} = 1.4 \text{ m s}^{-1}$
Recall the equation for kinetic energy.	$E_k = \frac{1}{2} mv^2$
Substitute the values for this situation into the equation.	$E_k = \frac{1}{2} \times 80 \times 1.4^2$
State the answer with appropriate units.	$E_k = 78 \text{ J}$

Worked example: Try yourself 9.2.2
CALCULATING KINETIC ENERGY CHANGES

As a bus with a mass of 10 tonnes approaches a school it slows from 60 km h^{-1} to 40 km h^{-1} .

a Calculate the work done by the brakes in the bus. Give your answers correct to two significant figures.	
Thinking	Working
Convert the values into SI units.	$u = 60 \text{ km h}^{-1}$ $= \frac{60 \text{ km}}{1 \text{ h}}$ $= \frac{60\,000 \text{ m}}{3600 \text{ s}}$ $= 17 \text{ m s}^{-1}$ $v = 40 \text{ km h}^{-1}$ $= \frac{40 \text{ km}}{1 \text{ h}}$ $= \frac{40\,000 \text{ m}}{3600 \text{ s}}$ $= 11 \text{ m s}^{-1}$ $m = 10 \text{ tonnes}$ $= 10\,000 \text{ kg}$
Recall the work–energy theorem.	$W = \frac{1}{2} \times mv^2 - \frac{1}{2} \times mu^2$
Substitute the values for this situation into the equation.	$W = \frac{1}{2}(10\,000 \times 11^2) - \frac{1}{2}(10\,000 \times 17^2)$
State the answer with appropriate units.	$W = -840\,000 \text{ J} = -840 \text{ kJ}$ Note: the negative value indicates that the work has caused the kinetic energy to decrease.
b The bus travels 40 m as it decelerates. Calculate the average force applied by the truck's brakes.	
Thinking	Working
Recall the definition of work.	$W = Fs$
Substitute the values for this situation into the equation. Note: The negative has been ignored as work is a scalar.	$840\,000 \text{ J} = F \times 40 \text{ m}$
Transpose the equation to find the answer.	$F = \frac{W}{s} = \frac{840\,000 \text{ J}}{40 \text{ m}} = 21\,000 \text{ N}$

Worked example: Try yourself 9.2.3
CALCULATING SPEED FROM KINETIC ENERGY

A 300 kg motorbike has 150 kJ of kinetic energy. Calculate the speed of the motorbike in km h^{-1} . Give your answer correct to two significant figures.	
Thinking	Working
Recall the equation for kinetic energy.	$E_k = \frac{1}{2} mv^2$
Transpose the equation to make v the subject.	$v = \sqrt{\frac{2E_k}{m}}$
Substitute the values for this situation into the equation.	$v = \sqrt{\frac{2 \times 150\,000}{300}} = 31.6 \text{ m s}^{-1}$
State the answer with appropriate units.	$v = 31.6 \times 3.6 = 114 \text{ km h}^{-1}$

Section 9.2 Review

KEY QUESTIONS SOLUTIONS

1 $80 \text{ km h}^{-1} = 22.22 \text{ m s}^{-1}$

$$E_k = \frac{1}{2}mv^2$$

$$= \frac{1}{2}230 \times 22.22^2 = 56\,779 \text{ J or } 57 \text{ kJ}$$

2 $W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2} \times 1500 \times 28^2 - \frac{1}{2} \times 1500 \times 17^2 = 370\,000 \text{ J}$

3 $v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 5000}{(72 + 9)}} = 11 \text{ m s}^{-1} = 40 \text{ km h}^{-1}$

4 $E_k = \frac{1}{2}mv^2 \therefore E_{k \propto} m$

So doubling the mass causes E_k to increase by a factor of 2 as well.

5

If $E_{k1} = \frac{1}{2}mv_1^2$ then

$$E_{k2} = \frac{1}{2}m(v_2)^2$$

$$= \frac{1}{2}m(3v_1)^2$$

$$= 9 \times \frac{1}{2}m(v_1)^2$$

$$= 9 \times E_{k1}$$

Hence the kinetic energy of the object is increased by a factor of 9.

6 The work-energy theorem defines work as the change in kinetic energy, so Lauren needs to increase her kinetic energy by 1000 J.

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = 1000$$

$$\frac{1}{2} \times 57.0 \times (v^2 - 0.500^2) = 1000$$

$$v^2 - 0.500^2 = 35.09$$

$$v^2 = 35.34$$

$$\text{So } v = 5.94 \text{ m s}^{-1}$$

7 $E_k = \frac{1}{2}mv^2$

Convert km h^{-1} to m s^{-1} :

For car 1 (travelling at 65 km h^{-1})

$$65.0 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{\text{km}} \times \frac{\text{h}}{3600 \text{ s}}$$

$$= \frac{65.0}{3.60}$$

$$= 18.06 \text{ m s}^{-1}$$

For car 2 (travelling at 60 km h^{-1})

$$\frac{60.0}{3.60} = 16.67 \text{ m s}^{-1}$$

car 1:

$$E_{k1} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 1800 \times 18.06^2$$

$$= 2.94 \times 10^5 \text{ J}$$

car 2:

$$\begin{aligned}
 E_{k1} &= \frac{1}{2}mv^2 \\
 &= \frac{1}{2} \times 1800 \times 16.67^2 \\
 &= 2.50 \times 10^5 \text{ J}
 \end{aligned}$$

i.e. car 1 has an extra 44 000 J of energy just by going 5 km h⁻¹ faster. Energy differences such as these become very significant in the event of a collision.

Section 9.3 Elastic and inelastic collisions

Worked example: Try yourself 9.3.1

ELASTIC OR INELASTIC COLLISION?

A 200g snooker ball with initial velocity 9.0 ms ⁻¹ to the right collides with a stationary snooker ball of mass 100g. After the collision, both balls are moving to the right and the 200g ball has a speed of 3.0 ms ⁻¹ . Show calculations to test whether or not the collision is inelastic.	
Thinking	Working
Use conservation of momentum to find the final velocity of the 100g ball.	Taking to the right as positive: $p_{i100} + p_{i200} = p_{f100} + p_{f200}$ $mv_{i100} + mv_{i200} = mv_{f100} + mv_{f200}$ $(0.1 \times 0) + (0.2 \times 9.0) = (0.1)v_{f100} + (0.2 \times 3.0)$ $v_{f100} = \frac{0 + 1.8 - 0.6}{0.1}$ $v_{f100} = 12 \text{ ms}^{-1}$
Calculate the initial kinetic energy before the collision.	$E_{ki200} = \frac{1}{2}mv_i^2$ $= \frac{1}{2} \times (0.2) \times (9.0)^2$ $= 8.1 \text{ J}$ $E_{ki100} = \frac{1}{2}mv_i^2$ $= \frac{1}{2} \times (0.1) \times (0)^2$ $= 0 \text{ J}$ $E_{ki} = E_{ki100} + E_{ki200}$ $= (0) + (8.1)$ $= 8.1 \text{ J}$
Calculate the final kinetic energy of the balls after the collision.	$E_{kf} = E_{kf100} + E_{kf200}$ $= \frac{1}{2}mv_f^2 + \frac{1}{2}mv_f^2$ $= \frac{1}{2} \times (0.1) \times (12)^2 + \frac{1}{2} \times (0.2) \times (3.0)^2$ $= 7.2 + 0.9$ $= 8.1 \text{ J}$
Compare the kinetic energy before and after the collision to determine whether or not the collision is inelastic.	The kinetic energy after the collision is the same as the kinetic energy before the collision. The collision is perfectly elastic.

Section 9.3 Review

KEY QUESTIONS SOLUTIONS

- 1 An elastic collision is one in which the kinetic energy of the objects involved before the collision is exactly equal to the kinetic energy of the objects after the collision. An inelastic collision is one where the total kinetic energy of the objects after the collision is lower. The law of conservation of energy places no restrictions on which type of energy is present; even though kinetic energy is lower, overall energy is still conserved.
- 2
 - a Take east as positive

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_3$$

$$1500 \times 12.0 + 2500 \times (-16.0) = (1500 + 2500) v_3$$

$$-22000 = 4000 v_3$$

$$v_3 = -5.50 \text{ ms}^{-1}$$
 So 5.50 ms^{-1} west
 - b Initial kinetic energy = $\frac{1}{2} \times 1500 \times 12.0^2 + \frac{1}{2} \times 2500 \times 16.0^2$

$$= 428000 \text{ J}$$
 Final kinetic energy = $\frac{1}{2} \times 4000 \times 5.50^2$

$$= 60500 \text{ J}$$
 Since the final kinetic energy is less than the initial kinetic energy, the collision is inelastic.
- 3

$$mv_1 = (m + m + m + m + m) v_2$$

$$0.0400 \times 1.50 = 5 \times 0.0400 \times v_2$$

$$v_2 = 0.300 \text{ ms}^{-1}$$
 Initial kinetic energy = $\frac{1}{2} \times 0.0400 \times 1.50^2$

$$= 0.0450 \text{ J}$$
 Final kinetic energy = $\frac{1}{2} \times 0.2 \times 0.300^2$

$$= 0.00900 \text{ J}$$
 So the collision is inelastic
- 4

$$p_i(\text{truck}) + p_i(\text{car}) = p_f(\text{truck}) + p_f(\text{car})$$

$$0.20 \times 0.30 + 0.10 \times 0.20 = 0.20 v_f + 0.10 \times 0.30$$

$$0.08 = 0.2 v_f + 0.03$$

$$0.2 v_f = 0.05$$

$$v_f = 0.05 \div 0.2$$

$$= 0.25 \text{ ms}^{-1}$$
- 5

$$E_{ki} = \frac{1}{2} m_t (v_{ti})^2 + \frac{1}{2} m_c (v_{ci})^2$$

$$= \frac{1}{2} \times 0.200 \times 0.300^2 + \frac{1}{2} \times 0.100 \times 0.200^2$$

$$= 0.009 + 0.002$$

$$= 1.10 \times 10^{-2} \text{ J}$$
- 6

$$E_{kf} = \frac{1}{2} m_t (v_{tf})^2 + \frac{1}{2} m_c (v_{cf})^2$$

$$= \frac{1}{2} \times 0.200 \times 0.250^2 + \frac{1}{2} \times 0.100 \times 0.300^2 \text{ J}$$

$$= 6.25 \times 10^{-3} + 4.50 \times 10^{-3}$$

$$= 1.08 \times 10^{-2} \text{ J}$$
- 7
 - a The total kinetic energy before the collision is **more than** the total kinetic energy after the collision.
 - b The kinetic energy of the system of toys **is not** conserved.
 - c The total energy of the system of toys **is** conserved.
 - d The total momentum of the system of toys **is** conserved.
 - e The collision **is not** perfectly elastic because **kinetic energy** is not conserved.

Section 9.4 Gravitational potential energy

Worked example: Try yourself 9.4.1

CALCULATING GRAVITATIONAL POTENTIAL ENERGY

A person doing their grocery shopping lifts a 5 kg grocery bag to a height of 30 cm. Calculate the gravitational potential energy of the grocery bag at this height. Give your answer correct to two significant figures.	
Thinking	Working
Recall the formula for gravitational potential energy.	$E_g = mg\Delta h$
Substitute the values for this situation into the equation.	$E_g = 5 \times 9.80 \times 0.3$
State the answer with appropriate units and significant figures.	$E_g = 15 \text{ J}$

Worked example: Try yourself 9.4.2

CALCULATING GRAVITATIONAL POTENTIAL ENERGY RELATIVE TO A REFERENCE LEVEL

A father picks up his baby from its bed. The baby has a mass of 6.0 kg and the mattress of the bed is 70 cm above the ground. When the father holds the baby in his arms, it is 125 cm off the ground. Calculate the increase in gravitational potential energy of the baby, taking g as 9.80 N kg^{-1} and giving your answer correct to two significant figures.	
Thinking	Working
Recall the formula for gravitational potential energy.	$E_g = mg\Delta h$
Identify the relevant values for this situation. Subtract the baby's original position off the ground from its final position.	$m = 6 \text{ kg}$ $g = 9.80 \text{ N kg}^{-1}$ $\Delta h = 125 - 70 = 55 \text{ cm} = 0.55 \text{ m}$
Substitute the values for this situation into the equation.	$E_g = 6 \times 9.80 \times 0.55$
State the answer with appropriate units and significant figures.	$E_g = 32 \text{ J}$

Section 9.4 Review

KEY QUESTIONS SOLUTIONS

1 a $E_g = mg\Delta h = 0.057 \times 9.80 \times 8.2 = 4.6 \text{ J}$

b $E_g = mg\Delta h = 0.057 \times 9.80 \times 4.1 = 2.3 \text{ J}$

2 $E_g = mg\Delta h$

$= 65 \times 9.80 \times (8848 - 5150)$

$= 2.36 \times 10^6 \text{ J}$

$= 2360 \text{ kJ}$

3 $E_g = mg\Delta h$

$20000 = 90.0 \times g \times 60.0$

$g = \frac{20000}{90.0 \times 60.0}$

$= 3.70 \text{ m s}^{-2}$

4 $E_g = mg\Delta h$

$= 50.0 \times 9.80 \times (1.80 - 0.75)$

$= 514.5 \text{ J}$

So Isabella has jumped with more energy than is required to equal the change in gravitational potential energy. She will therefore clear the bar.

$$\begin{aligned}
 5 \quad E_g &= mg\Delta h \\
 &= 7.50 \times 9.80 \times (-150) \\
 &= -11025 \text{ J}
 \end{aligned}$$

So the eagle's potential energy has decreased by $1.10 \times 10^3 \text{ J}$.

- 6 No. In physics, work against gravity is defined as force exerted over a displacement. Holding the weight above your head might require effort and energy, but you will not be doing any actual work against gravity.

Section 9.5 Law of conservation of energy

Worked example: Try yourself 9.5.1

MECHANICAL ENERGY OF A FALLING OBJECT

A 6.8 kg bowling ball is dropped from a height of 0.75 m. Calculate its kinetic energy at the instant before it hits the ground.	
Thinking	Working
Since the ball is dropped, its initial kinetic energy is zero.	$(E_k)_{\text{initial}} = 0 \text{ J}$
Calculate the initial gravitational potential energy of the ball.	$ \begin{aligned} (E_g)_{\text{initial}} &= mgh \\ &= 6.8 \times 9.80 \times 0.75 \\ &= 50 \text{ J} \end{aligned} $
Calculate the initial mechanical energy.	$ \begin{aligned} (E_m)_{\text{initial}} &= (E_k)_{\text{initial}} + (E_g)_{\text{initial}} \\ &= 0 + 50 \\ &= 50 \text{ J} \end{aligned} $
At the instant the ball hits the ground, its gravitational potential energy is zero.	$(E_g)_{\text{final}} = 0 \text{ J}$
Mechanical energy is conserved in this situation.	$ \begin{aligned} \therefore (E_m)_{\text{initial}} &= (E_m)_{\text{final}} = 50 \\ &= (E_k)_{\text{final}} + 0 \\ (E_k)_{\text{final}} &= 50 \text{ J} \end{aligned} $

Worked example: Try yourself 9.5.2

FINAL VELOCITY OF A FALLING OBJECT

A 6.8 kg bowling ball is dropped from a height of 0.75 m. Calculate its speed at the instant before it hits the ground.	
Thinking	Working
Recall the velocity of the falling object formula.	$v = \sqrt{2gh}$
Substitute the relevant values into the formula and solve.	$ \begin{aligned} v &= \sqrt{2 \times 9.80 \times 0.75} \\ &= 3.8 \text{ m s}^{-1} \end{aligned} $
Interpret the answer.	The bowling ball will be falling at 3.8 m s^{-1} just before it hits the ground.

Worked example: Try yourself 9.5.3
USING MECHANICAL ENERGY TO ANALYSE PROJECTILE MOTION

An arrow with a mass of 35 g is fired into the air at 80 ms^{-1} from a height of 1.4 m. Calculate the speed of the arrow when it has reached a height of 30 m.	
Thinking	Working
Recall the formula for mechanical energy.	$E_m = E_k + E_g = \frac{1}{2}mv^2 + mgh$
Substitute in the values for the arrow as it is fired.	$\begin{aligned}(E_m)_{\text{initial}} &= (E_k)_{\text{initial}} + (E_g)_{\text{initial}} \\ &= \frac{1}{2}mv^2 + mgh \\ &= \frac{1}{2}(0.035 \times 80^2) + (0.035 \times 9.80 \times 1.4) \\ &= 112.5 \text{ J}\end{aligned}$
Use conservation of mechanical energy to find an equation for the final speed.	$\begin{aligned}(E_m)_{\text{final}} &= (E_k)_{\text{final}} + (E_g)_{\text{final}} \\ &= \frac{1}{2}mv^2 + mgh \\ 112.5 &= \frac{1}{2}(0.035)v^2 + (0.035 \times 9.80 \times 30)\end{aligned}$
Solve the equation algebraically to find the final speed.	$\begin{aligned}112.5 &= 0.0175v^2 + 10.3 \\ 102.2 &= 0.0175v^2 \\ v^2 &= \frac{102.2}{0.0175} \\ v &= \sqrt{5840} \\ &= 76 \text{ ms}^{-1}\end{aligned}$
Interpret the answer.	The arrow will be moving at 76 ms^{-1} when it reaches a height of 30 m.

Worked example: Try yourself 9.5.4
ENERGY EFFICIENCY

An electric kettle uses 23.3 kJ of electrical energy as it boils a quantity of water. The efficiency of the kettle is 18%.

How much electrical energy is expended in actually boiling the water?	
Thinking	Working
Recall the equation for efficiency. Substitute the given values into the equation.	$\begin{aligned}\text{efficiency} &= 18\% \\ \text{input} &= 23.3 \text{ kJ} \\ \text{output} &= ? \\ \text{efficiency } (\eta) &= \frac{\text{energy output}}{\text{energy input}} \times 100\% \\ 18 &= \frac{\text{energy output}}{23.3} \times 100\%\end{aligned}$
Solve the equation.	output = 4.19 kJ

Section 9.5 Review

KEY QUESTIONS SOLUTIONS

1 a $(E_k)_{\text{final}} = (E_g)_{\text{initial}} = mg\Delta h = 180 \times 9.80 \times 15 = 26\,460\text{ J}$

To two significant figures: 26 000 J

b $(E_g)_{\text{initial}} = +(E_k)_{\text{final}}$

$$26\,460 = (E_k)_{\text{final}} + mg\Delta h$$

$$26\,460 = (E_k)_{\text{final}} + (180 \times 9.80 \times 5)$$

$$26\,460 - 8820 = (E_k)_{\text{final}}$$

$$(E_k)_{\text{final}} = 17\,640\text{ J}$$

2 a $E_{\text{initial}} = E_{\text{final}}$

$$\frac{1}{2}mv^2 = mg\Delta h$$

$$v = \sqrt{2gh} = \sqrt{2 \times 9.80 \times 15} = 17\text{ ms}^{-1}$$

b $v = \sqrt{2gh} = \sqrt{2 \times 9.80 \times 10} = 14\text{ ms}^{-1}$

3 $v = \sqrt{2gh}$

$$h = \frac{v^2}{2g} = \frac{5.4^2}{2 \times 9.80} = 1.5\text{ m}$$

4 a $E_m = E_k + E_g$
 $= mv^2 + mgh$

$$= \frac{1}{2} \times 0.800 \times 28.5^2 + 0.800 \times 9.80 \times 1.45$$

$$= 336\text{ J}$$

b $E_m = E_k + E_g$

$$336 = \frac{1}{2} \times 0.800 \times v^2 + 0$$

$$v = \sqrt{\frac{336}{0.400}}$$

$$= 29.0\text{ ms}^{-1}$$

5 Energy output = $\frac{\text{efficiency}}{100} \times \text{energy input}$

$$= \frac{30}{100} \times 2000 = 600\text{ J}$$

6 80% of its E_g is retained so 80% of its height is retained. 80% of 1.5 m = 1.2 m

7 The ball is moving upwards, hence its gravitational potential energy is increasing. By the conservation of energy, its kinetic energy must therefore be decreasing by the exact same amount.

Section 9.6 Power

Worked example: Try yourself 9.6.1

CALCULATING POWER

Calculate the power used by a weightlifter to lift a barbell which has a total mass of 50 kg from the floor to a height of 2.0 m above the ground in 1.4 s. (Use $g = 9.80 \text{ m s}^{-2}$.)	
Thinking	Working
Calculate the force applied.	$F_g = mg$ $= 50 \times 9.80$ $= 490 \text{ N}$
Calculate the work done.	$W = F_s$ $= 490 \times 2.0$ $= 980 \text{ J}$
Recall the formula for power.	$P = \frac{W}{\Delta t} \text{ or } P = \frac{\Delta E}{\Delta t}$
Substitute the appropriate values into the formula.	$P = \frac{980}{1.4}$
Solve.	$P = 700 \text{ W}$

Worked example: Try yourself 9.6.2

FORCE-VELOCITY FORMULATION OF POWER

Calculate the power required to keep a car moving at an average speed of 22 m s^{-1} if the force of friction (including air resistance) is 1200 N. Give your answer correct to three significant figures.	
Thinking	Working
Recall the force-velocity formulation of the power equation.	$P = Fv_{av}$
Substitute the appropriate values into the formula.	$P = 1200 \times 22$
Solve.	$P = 26\,400 \text{ W}$

Section 9.6 Review

KEY QUESTIONS SOLUTIONS

1 $100 \text{ km h}^{-1} \div 3.6 = 27.8 \text{ m s}^{-1}$

$$\begin{aligned}
 P &= \frac{\Delta E}{\Delta t} \\
 &= \frac{\frac{1}{2}(1610 \times 27.8^2)}{5.50} \\
 &= 113\,115 \text{ W or } 113 \text{ kW}
 \end{aligned}$$

2 $P = 4000 \times 20$
 $= 80\,000 \text{ W}$
 $= 80 \text{ kW}$

3 $80 \text{ km h}^{-1} \div 3.6 = 22.2 \text{ m s}^{-1}$

$$\begin{aligned}
 P &= Fv_{av} \\
 \therefore F &= \frac{P}{v_{av}} \\
 &= \frac{40\,000}{22} \\
 &= 1800 \text{ N}
 \end{aligned}$$

$$4 \quad P = Fv_{av}$$

$$\therefore v_{av} = \frac{P}{F}$$

$$F = mg = 500 \times 9.80 = 4900 \text{ N}$$

$$v_{av} = \frac{P}{F} = \frac{25\,000}{4900} = 5.1 \text{ ms}^{-1}$$

$$5 \quad v_{av} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{20.0}{10.0}$$

$$= 2.00 \text{ ms}^{-1}$$

$$P = Fv_{av}$$

$$= 15.0 \times 2.00$$

$$v_{av} = 30.0 \text{ W}$$

$$6 \quad P = \frac{W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

$$W = mg\Delta h$$

$$= 40.0 \times 9.80 \times 1.50$$

$$= 588 \text{ J}$$

$$\therefore P = \frac{588}{10.0}$$

$$= 58.8 \text{ W}$$

$$7 \quad P = \frac{W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

$$\Delta E = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$= \frac{1}{2} \times 800.0 \times (55.0^2 - 40.0^2)$$

$$= 5.70 \times 10^5 \text{ J}$$

$$\therefore 120 \times 10^3 = \frac{5.70 \times 10^5}{\Delta t}$$

$$\therefore \Delta t = \frac{5.70 \times 10^5}{120 \times 10^3}$$

$$= 4.75 \text{ s}$$

CHAPTER 9 REVIEW

- 1 $W = Fs = 2000 \times 80 = 160\,000 \text{ J}$
- 2 Approximately 40 squares, so 40 J
- 3 $W = Fs$
 $= mg \times h$
 $= 200 \times 9.80 \times 30$
 $= 58\,800 \text{ J}$
- 4 For each step: $W = Fs = 60 \times 9.80 \times 0.165 = 97 \text{ J}$
 For all 12 steps: $W = 12 \times 97 = 1200 \text{ J}$
- 5 $F = mg = 50 \times 9.80 = 490 \text{ N}$
 $s = \frac{W}{F} = \frac{4000}{490} = 8.2 \text{ m}$
- 6 $W = Fs \cos \theta$
 $1200 = F \times 20 \cos 35^\circ$
 $F = 73.25 \text{ N}$

7 $150 \text{ km h}^{-1} = 42 \text{ m s}^{-1}$

$160 \text{ g} = 0.16 \text{ kg}$

$$E_k = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 0.16 \times 42^2 = 140 \text{ J}$$

8 $v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 70\,000}{1200}} = 11 \text{ m s}^{-1}$

9 $E_k = \frac{1}{2}mv^2 \therefore E_k \propto v^2$

So doubling the velocity causes E_k to change by a factor of 2^2 or 4.

10 $E_g = mg\Delta h = 88 \times 9.80 \times 0.40 = 340 \text{ J}$

11 $E_m = E_k + E_g$

$$= \frac{1}{2}mv^2 + mgh$$

$$= \frac{1}{2} \times (0.43 \times 16^2) + (0.43 \times 9.80 \times 0)$$

$$= 55 \text{ J}$$

$$E_m = E_k + E_g$$

$$55 = \frac{1}{2} \times (0.43 \times v^2) + (0.43 \times 9.80 \times 2.44)$$

$$= 0.215v^2 + 10.3$$

$$v = \sqrt{\frac{44.7}{0.215}}$$

$$= 14.4 \text{ m s}^{-1}$$

12 Initial kinetic energy is contained only within the white ball.

So

$$E_k = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 0.160 \times 5.00^2$$

$$= 2.00 \text{ J}$$

$$\text{Final kinetic energy} = \frac{1}{2} \times 0.160 \times 1.00^2 + \frac{1}{2} \times 0.160 \times 4.00^2$$

$$= 1.36 \text{ J}$$

The final kinetic energy of the system is lower, hence the collision was inelastic.

13 a $E_g = mgh$

$$= 1.51 \times 9.80 \times 0.15 = 2.2 \text{ J}$$

b The gain in gravitational potential energy of the pendulum (2.2 J) is equal to the kinetic energy of the pendulum as it starts to swing upwards, so the pendulum had 2.2 J of kinetic energy.

c $v = \sqrt{\frac{2E_k}{m}}$

$$= \sqrt{(2 \times 2.2 \div 1.51)}$$

$$= 1.7 \text{ m s}^{-1}$$

14 Remember: $P = \frac{W}{\Delta t}$ and $1 \text{ kW} = 1000 \text{ W}$

$$P = \frac{mgh}{\Delta t}$$

$$= \frac{5000 \times 9.80 \times 20}{5}$$

$$= 196\,000 \text{ W}$$

$$= 196 \text{ kW} \approx 200 \text{ kW}$$

$$\begin{aligned}
 \mathbf{15} \quad \Delta E_k &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \\
 &= \frac{1}{2} \times (650 \times 27.8^2) - 0 \\
 &= 251\,000\text{ J} \\
 &= 251\text{ kJ}
 \end{aligned}$$

$$\begin{aligned}
 P &= \frac{\Delta E}{t} \\
 &= \frac{251}{7.2} \\
 &= 35\text{ kW}
 \end{aligned}$$

$$\mathbf{16} \quad F = \frac{P}{V_{av}} = \frac{25\,000}{17} = 1500\text{ N}$$

$$\mathbf{17} \quad \mathbf{a} \quad W = E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 60 \times 8^2 = 1920\text{ J}$$

$$\mathbf{b} \quad F = \frac{W}{s} = \frac{1920}{20} = 96\text{ N}$$

$$\mathbf{18} \quad E_g = mg\Delta h = 120 \times 1.6 \times 0.1 = 19.2\text{ J}$$

$$\mathbf{19} \quad \text{Efficiency } (\eta) = \frac{\text{energy output}}{\text{energy input}} \times 100\%$$

$$80 = \frac{1250}{\text{input}} \times 100$$

$$\begin{aligned}
 \text{input} &= \frac{1250}{80} \times 100 \\
 &= 1562.5\text{ J}
 \end{aligned}$$